Title: E-commerce Return Rate Analysis

1️⃣ Introduction

E-commerce has become one of the fastest-growing industries in the world. Customers can easily purchase products online from anywhere at any time. However, one major challenge faced by e-commerce companies is product returns.

Product returns increase operational costs, reduce profits, and affect customer satisfaction. Understanding the reasons for returns and identifying patterns can help businesses reduce losses and improve service quality.

This project focuses on analyzing e-commerce order data to study return behavior and identify factors influencing product returns.

2️⃣ Objectives

The main objectives of this project are:

- To analyze product return patterns in e-commerce data
- To identify categories with high return rates
- To analyze state-wise return trends
- To study the relationship between price, freight charges, and returns
- To build a machine learning model to predict return probability

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3️⃣ Dataset Description

The dataset used for this project is the **Brazilian E-Commerce Public Dataset by Olist** from Kaggle.

It contains real-world e-commerce transaction data of approximately 100,000 orders.

📊 Files Used:

- olist_orders_dataset.csv
- olist_order_items_dataset.csv
- olist_products_dataset.csv
- olist_customers_dataset.csv
- olist_order_reviews_dataset.csv
- olist_order_payments_dataset.csv

4️⃣ Tools & Technologies Used

- Google Colab
- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- GitHub

5️⃣ Data Cleaning & Preprocessing

The following preprocessing steps were performed:

- Imported all required CSV files
- Checked missing/null values
- Removed duplicate records
- Merged multiple tables using common keys
- Created a new column named **Returned**

Return Logic:

Orders with status:

- canceled
- unavailable

were considered as returned orders.

6️⃣ Exploratory Data Analysis (EDA)

EDA was performed to understand return patterns.

Key Analysis:

- Overall return rate
- Return rate by product category
- Return rate by customer state
- Price vs return comparison
- Freight charges vs return trend

7️⃣ Return Rate Calculation

Return Rate was calculated using:

Return Rate = (Returned Orders / Total Orders) × 100

This helped measure the percentage of problematic orders.

8️⃣ Data Visualization

The following charts were created:

- Bar chart: Top categories with highest returns
- Bar chart: States with highest returns
- Box plot: Price vs Return
- Trend charts for insights

Visualizations made it easier to understand patterns clearly.

9️⃣ Machine Learning Model

A classification model was built to predict whether an order would be returned.

Model Used:

- Logistic Regression

Input Features:

- Product Price
- Freight Value

Target Variable:

- Returned (1 = Yes, 0 = No)

🔟 Model Evaluation

The model performance was measured using:

- Accuracy Score
- Prediction Results

The model successfully predicted return behavior based on available features.

1️⃣1️⃣ Key Insights

- Some product categories have higher return rates than others

- Certain states showed more returned orders
- Higher freight charges may influence returns
- Cancelled and unavailable orders significantly impact return rate
- Predictive analytics can help reduce future returns

1️⃣2️⃣ Business Recommendations

Based on analysis:

- Improve stock management to avoid unavailable orders
- Improve delivery process
- Focus on high-return product categories
- Optimize shipping costs
- Use prediction model to identify risky orders early

1️⃣3️⃣ Conclusion

This project successfully analyzed e-commerce return data using Python and machine learning techniques. Important return patterns were identified across categories and customer locations.

The predictive model demonstrated how businesses can proactively reduce returns and improve profitability. This project provided practical experience in data cleaning, analysis, visualization, and machine learning using real-world data.

1️⃣4️⃣ Future Scope

- Use advanced models like Random Forest or XGBoost
- Include customer review sentiment analysis
- Build Power BI dashboard
- Real-time return monitoring system

1️⃣5️⃣ References

- Kaggle Olist Dataset
- Python Documentation
- Scikit-learn Documentation